FLEET TEAM DIGEST

737MAX-FTD-30-20001

Issue Title : Window Heat Faults

Airplane Model	ATA	Minor Model(s)
737MAX	3041-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
02/24/2020	11/14/2023	07/12/2023

Estimated Completion Date	Next Update Date

Revision Description

Status, Milestones, and References update.

Applicability

All 737MAX Airplanes

Description

The Window Heat Control Unit (WHCU), P/N 8300-27904 has been reported by some Operators to have window OVERHEAT Light illumination condition resulting in operation delays and WHCU removals/replacement.

Background

Various Operators have reported frequent Window OVERHEAT light illumination condition with Window Heat Control Unit (WHCU) P/N 83000-27904, resulting in operation delays WHCU removals, and shop No Fault Found (NFF) conditions. The WHCU is part of the control cabin window anti-icing system, and provides window temperature monitoring that supplies ON/OFF and OVERHEAT system indications.

WHCU P/N 83000-27904 is an improved version from previous WHCU P/N 83000-27901, which added functionality, weight improvements, and addressed anomalies related to the following:

1. Startup Anomaly: The WHCU goes through an initialization process every time the window heat system is activated, which could be mistaken for an abnormal indication.

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2. Potential Anomaly during Power Transfer: Since the WHCU was incorporated into production, there has been reports of the window OVERHEAT lights illuminating during bus transfers on the ground. In each case, the WHCU was reset following the QRH Non-Normal Procedure for WINDOW OVERHEAT. The fault shown on the WHCU BITE display for this condition is “WHCU-LRU”.
3. WHCU Relay Failure: A couple reports of WHCU hard failures has caused the window heat circuit breakers to trip. Investigation of the failed WHCUs revealed that the overcurrent protection relays within the WHCU failed. These relay failures prevented the WHCU from shutting down during overcurrent situations and resulted in WHCU internal component failures.

Although WHCU P/N 83000-27904 reduced/improved instances of the Overheat Light illumination condition, various Operators are still reporting window Overheat Light illumination condition with WHCU P/N 83000-27904.

Status

Boeing and the WHCU Supplier, Koito Manufacturing Co. Ltd., have conducted an investigation into the Window Overheat Light illumination condition being reported by Operators in the field.

Field reliability data from approximately 2 years shows about 33 Operators reported Window Overheat light condition. The top 5 Operators reported issue with approximately 15-25% of Fleet and remaining Operators reported issue with approximately 1-20% of Fleet. The Overall Operators average reported issue was approximately 9%.

Initial analysis of returned units indicated three main Faults were being recorded:

1. Temperature Sensor Faults
2. Overheat Faults
3. Command without Heat Fault

The Temperature Sensor Faults were determined to be attributed to sensor degradation/failures. The Overheat Faults were determined to be a misdetection of WHCU due to deterioration and failure of the window’s temperature sensor. The units that have been returned to Koito for repairs were further analyzed and confirmed to contain the Temperature Sensor Faults, the Overheat Faults, and the Command without Heat fault in the unit's Non-Volatile Memory (NVM). These units were also detemrined to be NFF, and removal could have been prevented with a reset by cycling the window heat power.

Boeing and Koito investigated a timing and noise issue (ie vibration) that was determined to be triggering the above three faults in the NVM, which is contributing to many of the NFF conditions. Boeing and Koito identified a resolution, which includes a software update that would allow for getting more sample readings than the current implementation prior to triggering a fault in the health monitoring logic.

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After various reviews and iterations, Koito completed testing the software updates for applicability and the final proposed software updates have been approved by Boeing. Koito provided Boeing with a draft Service Bulletin (SB) and after various iterations updates, Koito submitted the formal SB to Boeing. SB is currently in review process as Boeing and Koito negotiate industry support language. Note, a fielded WHCU will be required to go back to Supplier, Koito to perform Software update as the Software update cannot be performed on-wing. For production incorporation, WHCU SN 6211 assigned to Aircraft LN 8229. There will be no change to the current 83000-27904 part number. For configuration and tracking purposes, an updated WHCU will have a "Mod A" noted on the part number label of the WHCU.

Koito's Service Bulletin (SB 83000-30-061) and Boeing Service Letter (SL 737-SL-30-037) have been released. Any operator wishing to get a copy of Koito's SB must contact Koito directly (preferably via email) as Boeing is not authorized to provide a copy, per guideline. Below is Koito's contact information:

Koito Manufacturing Co. Ltd.

701 5th Ave. Suite 1160

Seattle, WA 98104 U.S.A.

TEL: 1-206-386-5624

Point of Contact:

Yukitoshi Inagaki
email: yinagaki@koito.co.jp

Interim Action

FCOM/QRH Non-Normal Checklist provides instructions to reset the Window Heat system by selecting the WINDOW HEAT switch to OFF then back ON in the event of an OVERHEAT light illumination.

IFIM 30-41-00-810-828, -829, -830, and -831 were revised to include instructions to perform a system reset if an OVERHEAT light illuminates. If the OVERHEAT light is no longer active after system reset, then the WHCU does not need to be replaced.

Service Letter 737-SL-30-028-D provides a maintenance recommendation to reduce the removals of the WHCU, P/N 83000-27904 due to Window OVERHEAT light illuminations during bus power transfers.

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Final Action

Koito will be implementing the latest software in new WHCUs, which will be identified as Mod A beginning with S/N 06211. The software update is not on-wing loadable. Koito will be releasing a supplier service bulletin. Boeing will issue a service letter notifying operators of the availability of the supplier service bulletin.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: COMPLETE

Solution Selected: COMPLETE

Change Committed: COMPLETE

Supplier Service Bulletin Available: COMPLETE

Service Letter Available: COMPLETE

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	737-SL-30-037		
Service Letter (SL)	737-SL-56-017		
Service Bulletin (SB)	83000-30-061		
Service Letter (SL)	737-SL-30-028	D	
Fleet Team Digest Article (FTD)	737NG-FTD-30-16002		
Fault Isolation Manual (FIM)	IFIM 30-41-00-810-828, -829, -830, and -831		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Window Heat Control Unit (WHCU)	10-61833-9	Koito	83000-27904

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Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board

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